

**B.Tech. Degree IV Semester Supplementary Examination
March 2024**

CE/CS/EC/EE/IT/ME/SE 19-200-0401

COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Transform a region to another using conformal mapping.

CO2: Evaluate real integral using Residue theorem.

CO3: Form and solve partial differential Equations.

CO4: Determine solution of Partial differential equation for vibrating string and heat conduction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) If $u = x^2 - y^2 - y$ is harmonic find the harmonic conjugate of u .	3	3	L2	1	2
(b) Show that the image of the hyperbola $x^2 - y^2 = 1$ under the transformation $w = \frac{1}{z}$ is the lemniscate $R^2 = \cos 2\phi$.	3	3	L2	1	2
(c) Evaluate $\int_C \frac{e^z dz}{(z-1)(z-4)}$ using Cauchy's integral formula where C is the circle $ z =2$.	3	3	L5	2	5
(d) Find the singularity of the function $f(z) = z^2 \frac{1}{e^z}$. Also find its type and residue at that singularity when $ z < 2$.	3	3	L2	2	4
(e) Solve: $py + qx + pq = 0$.	3	3	L2	3	2
(f) Form a partial differential by eliminating the function f from $z = f\left(\frac{y}{x}\right)$.	3	3	L2	3	2
(g) Derive the wave equation.	3	3	L1	4	1
(h) Solve by the method of separation of variables $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$.	3	3	L3	4	3

PART B

(4 × 12 = 48)

II. (a) If $f(z)$ is an analytic function with real part $u = e^{-2xy} \sin(x^2 - y^2)$ find $f(z)$ in terms of z .	6	6	L3	1	3
(b) Show that the bilinear transformation $w = \frac{5-4z}{4z-2}$ transform the circles $ z =1$ into a circles in the W plane and find its centre and radius. Also find the fixed points of the bilinear transformation.	6	6	L2	2	2

OR

(P.T.O.)

	Marks	BL	CO	PO
III. (a) Derive the polar form of Cauchy-Riemann equation and deduce that $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$.	6	L3	3	3
(b) Show that the mapping $w = z + \frac{1}{z}$ transforms the circle $ z =r$ into an ellipse in the w-plane. Also find the points in the z-plane where the transformation is not conformal.	6	L2	2	3
IV. (a) State and prove Cauchy's integral theorem.	4	L1	2	1
(b) Find the Laurent's expansion of $f(z) = \frac{z+3}{z(z^2-z-2)}$ in the region $1 < z < 2$ and hence find the residue of $f(z)$ at $z=0$.	4	L5	2	3
(c) Using Residue theorem evaluate $\oint_C \frac{e^z dz}{(z-1)^2(z-2)}$ where C is the circle $ z-1 =3$.	4	L5	2	3
OR				
V. (a) State and prove Cauchy's integral formula and using it evaluate $\oint_C \frac{dz}{z^2(z-3)}$ where C is the Circle $ z =1$.	7	L1, L5	2	1
(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{17-8\cos\theta}$ around the circle $ z =1$.	5	L5	2	3
VI. (a) Solve: $(y-z)p + (z-x)q = x-y$.	6	L3	3	3
(b) Solve: $(D^3 - 4D^2D' + 4DD'^2)z = \cos(2x+3y)$	6	L3	3	3
OR				
VII. (a) Solve: $y^2(x-y)p + x^2(y-x)q = z(x^2+y^2)$.	6	L3	3	3
(b) Solve: $r-s-2t=(y-1)e^x$.	6	L3	3	3
VIII. (a) Find the solution of the wave equation by the method of separation of variables.	6	L1	4	1
(b) A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin\left(\frac{\pi x}{l}\right)$ from which it is released at time $t=0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin y(x, t) = a \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{\pi ct}{l}\right)$.	6	L2	4	2
OR				
IX. (a) Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ by the method of separation of variables.	6	L1	4	1
(b) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x, 0) = 3 \sin(n\pi x)$, $u(0, t) = 0$ and $u(1, t) = 0$ where $0 < x < 1$, $t > 0$.	6	L2	4	2

Bloom's Taxonomy Levels

L₁ = 18%, L₂ = 38%, L₃ = 29%, L₅ = 15%